

Financial Instrument Similarity — Cheat Sheet

Goal: find instruments that are economically close, not merely similarly priced. Compare co-movement, unexplained differences, risk magnitude, and economic characteristics.

1. Core framework

Instrument A → returns/risk changes → regression against B → residual analysis → volatility comparison → rating/tenor/risk-factor checks → similarity score.

Feature	Meaning	Main use
Return correlation	Do they move together?	First screen
Beta	How much does A respond to B?	Sensitivity / hedge
Residuals	What B cannot explain	Closeness
Residual volatility	Size of unexplained movement	Key similarity measure
Cointegration	Stable long-run price relationship?	Pairs / tenor substitution
Volatility	Typical movement size	Risk magnitude
Volatility ratio	$\text{Vol}(A) \div \text{Vol}(B)$	Scale comparison
Rating	Credit quality	Bonds / CDS
Duration / DV01	Rate sensitivity	Bonds
CS01	Credit-spread sensitivity	CDS / credit
Greeks	Derivative sensitivities	Options / derivatives

2. Residual analysis

Regression example: $rA = \alpha + \beta rB + \varepsilon$. The residual ε is the part of A's return that B did not explain. If instruments are close, residuals are often small, centred near zero and stable. Large or changing residuals suggest different risk drivers.

Residual pattern	Interpretation	Similarity
Small random residuals around 0	Most movement explained	Strong positive evidence
Large but stable residuals	Related but different risk scale	Related, not equivalent
Trend in residuals	Missing factor / tenor effect	Weakens similarity
Runs above/below zero	Persistent divergence / regime	Investigate
Autocorrelation	Difference persists through time	Useful for spread/cointegration
Increasing variance	Relationship changes with risk regime	Not stable
Large spikes	Event, liquidity, bad data or break	Investigate

3. Which residual analysis is used?

Basic: residual mean and standard deviation. **Diagnostic:** residual-vs-fitted plot and time-series plot. **Time series:** autocorrelation and stationarity tests. **Production:** rolling residual volatility and stress-period residuals. For financial substitution, **residual volatility + stability over time + stress behaviour** are usually more useful than residual mean alone.

4. Volatility and volatility ratio

Volatility measures typical return movement. High correlation does not guarantee similar risk. **Volatility ratio = Vol(A) / Vol(B)**; a value near 1 means similar movement magnitude.

Pair	Correlation	Vol A	Vol B	Vol ratio	Reading
1	0.96	10%	10.5%	0.95	Very close
2	0.96	10%	20%	0.50	Move together but risk scale differs
3	0.60	10%	10.5%	0.95	Similar size, weaker co-movement
4	0.20	10%	30%	0.33	Poor substitute

5. Rating matters

For bonds and CDS, rating should normally be a **hard filter or major feature**. A 5Y BBB bond and a 5Y AA bond can have almost identical rate exposure while having very different credit risk. Compare rating together with spread, CS01, sector, seniority, issuer and liquidity.

A	B	Stats	Rating	Conclusion
5Y Corp	7Y Corp	Corr 0.97; low residual	BBB / BBB	Good tenor-substitute candidate
5Y Corp	5Y Corp	Corr 0.98; low residual	BBB / AA	Related, but credit quality differs
5Y CDS	5Y CDS	Corr 0.95; similar CS01	A / A	Strong candidate
Bank CDS	Energy CDS	Corr 0.90	A / BBB	Do not rely on correlation alone

6. Asset-class features

Asset	Key similarity features
Equity	Return correlation, beta, volatility, sector, country, factor exposures, stress beta
Equity derivatives	Delta, gamma, vega, theta, rho, moneyness, expiry, underlying/factor exposure
Bonds	Yield, spread, tenor, duration, DV01, convexity, rating, sector, seniority, liquidity
CDS	Spread, CS01, maturity, rating, hazard/default risk, sector, seniority, liquidity
FX	Returns, volatility, beta, carry/rate differential, forward behaviour, implied volatility

7. Tenor substitution example

If a 5Y bond is unavailable, do not select a 7Y bond just because price correlation is high. Compare **tenor + duration/DV01 + rating + spread + volatility + residual behaviour**. A 7Y bond with compatible credit quality, similar risk-normalised rate exposure and low stable residuals is a stronger substitute than a 10Y bond with only high correlation.

8. Practical similarity score

Component	Illustrative weight	Better result
Return correlation	20%	Higher
Normalised residual volatility	20%	Lower
Beta / sensitivity difference	15%	Closer
Volatility ratio difference from 1	10%	Closer to 1
Risk-factor distance	15%	Lower
Rating / credit compatibility	10%	Compatible
Tenor / duration / Greeks	10%	Closer

Weights must be calibrated by asset class and business purpose. Rating can be a hard eligibility constraint for credit substitution rather than merely a weighted score.

9. Residual plot cheat sheet

Visual pattern	Meaning
Random cloud around zero	Good linear relationship
Funnel / widening spread	Changing volatility; relationship depends on risk regime
Curve	Non-linear relationship is missing
Long runs above/below zero	Persistent divergence / regime effect
Big isolated spikes	Events, liquidity, bad data or structural break

10. Recommended decision tree

1. Apply economic filters: asset class, currency, issuer/underlying, rating, seniority, liquidity and allowable tenor. **2.** Calculate return/risk-change correlation and beta. **3.** Run regression and inspect residual size, plots and stability. **4.** Compare volatility and volatility ratio. **5.** Compare duration/DV01, CS01, Greeks and factor exposures. **6.** Test normal and stress regimes. **7.** Rank candidates.

GOLDEN RULE: High correlation ≠ identical risk. The strongest evidence of similarity is high co-movement + similar risk magnitude/sensitivity + small stable residuals + compatible economic characteristics such as rating, tenor, seniority and liquidity.